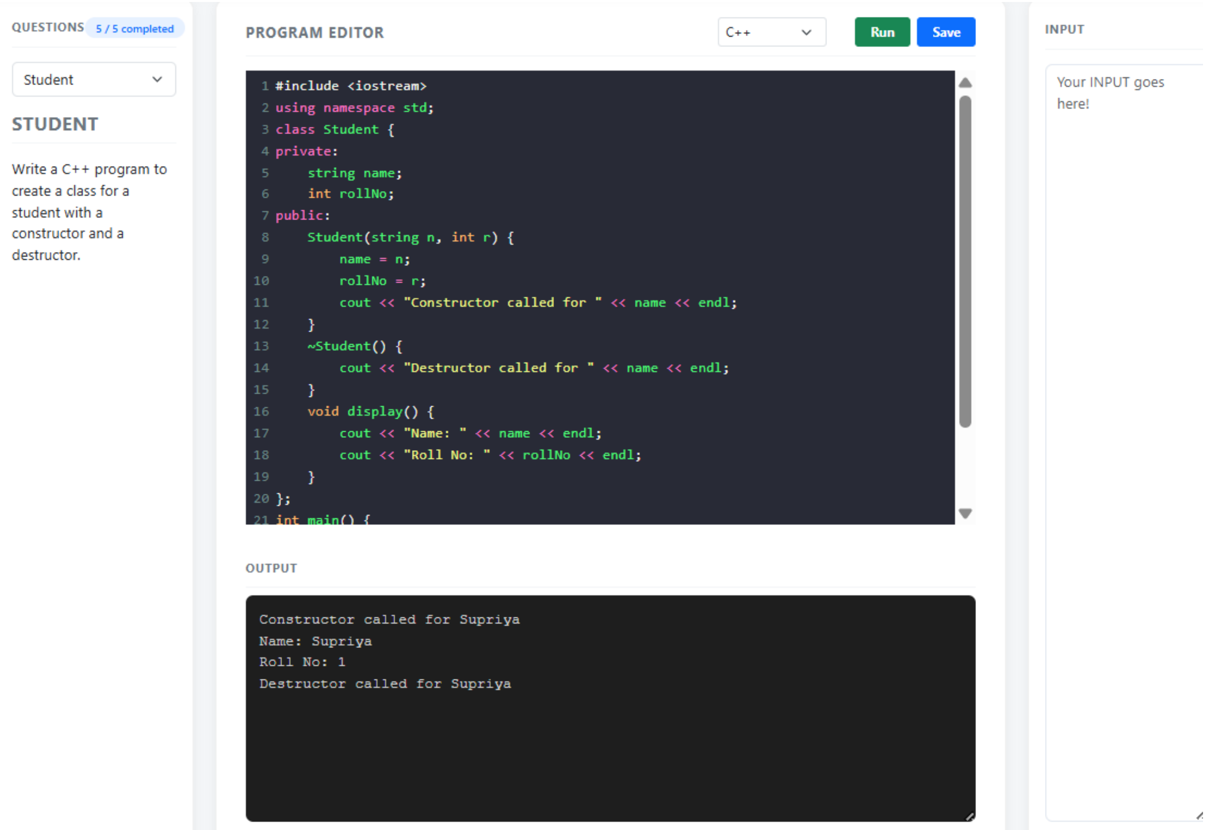
1.**Creating a class for a student with a constructor and destructor**

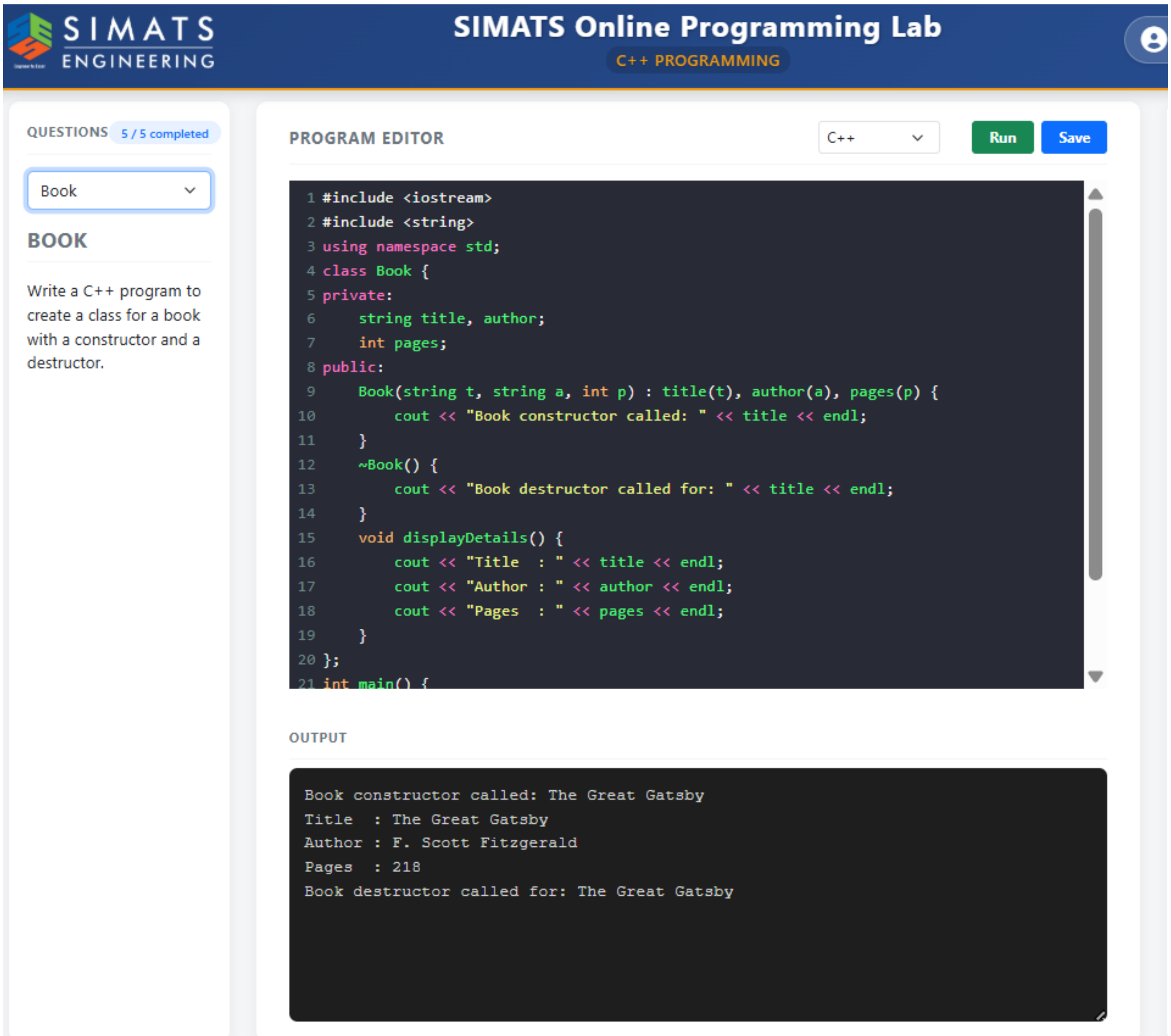
```
#include <iostream>
#include <string>
using namespace std;

class Student {
private:
string name;
int rollNo;

public:
// Constructor
Student(string n, int r) {
name = n;
rollNo = r;
cout << "Constructor called for " << name << endl;
}

// Destructor
~Student() {
cout << "Destructor called for " << name << endl;
}

// Display details
void display() {
cout << "Name: " << name << endl;
cout << "Roll No: " << rollNo << endl;
}
};

int main() {
Student s("Supriya", 1);
s.display();
return 0;
}
```



2.**Creating a class for a book with a constructor and destructor**

```
#include <iostream>
#include <string>
using namespace std;

class Book {
private:
string title, author;
int pages;

public:
Book(string t, string a, int p) : title(t), author(a), pages(p) {
cout << "Book constructor called: " << title << endl;
}

~Book() {
cout << "Book destructor called for: " << title << endl;
}

void displayDetails() {
cout << "Title : " << title << endl;
```

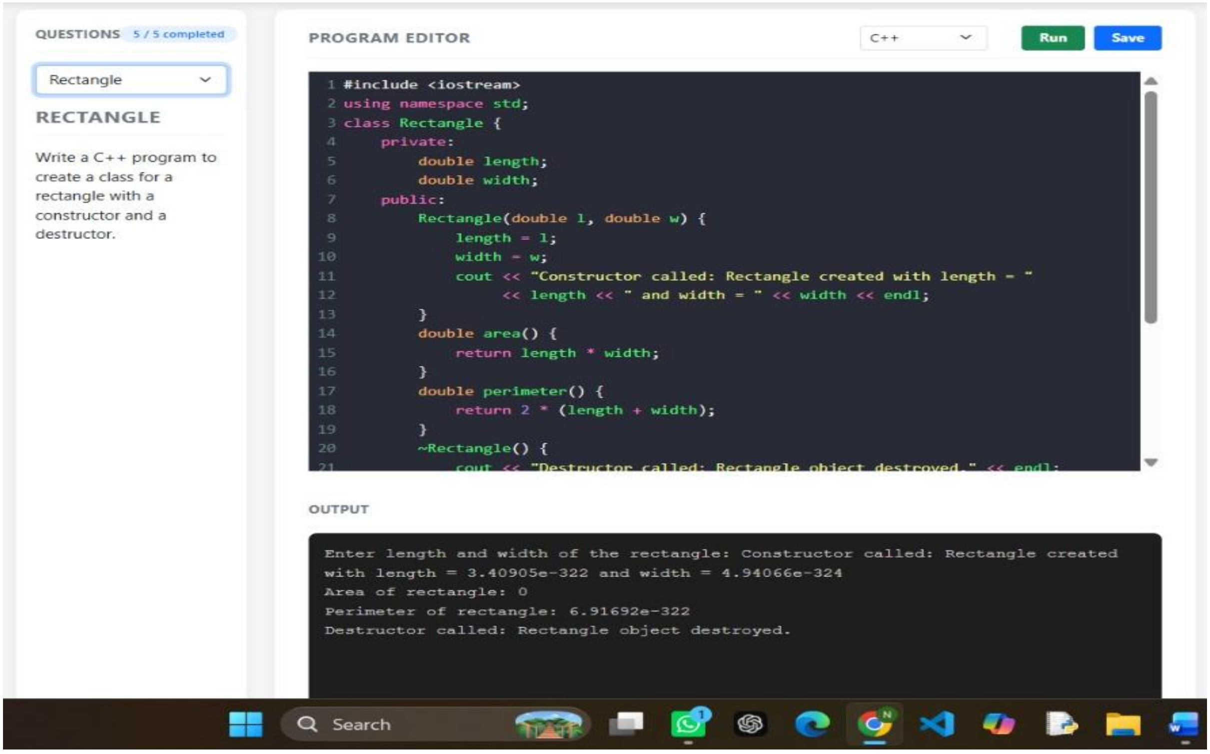
```
cout << "Author : " << author << endl;
cout << "Pages : " << pages << endl;
}
};

int main() {
Book b1("The Great Gatsby", "F. Scott Fitzgerald", 218);
b1.displayDetails();
return 0;
}
```



3.**Creating a class for a rectangle with a constructor and destructor**

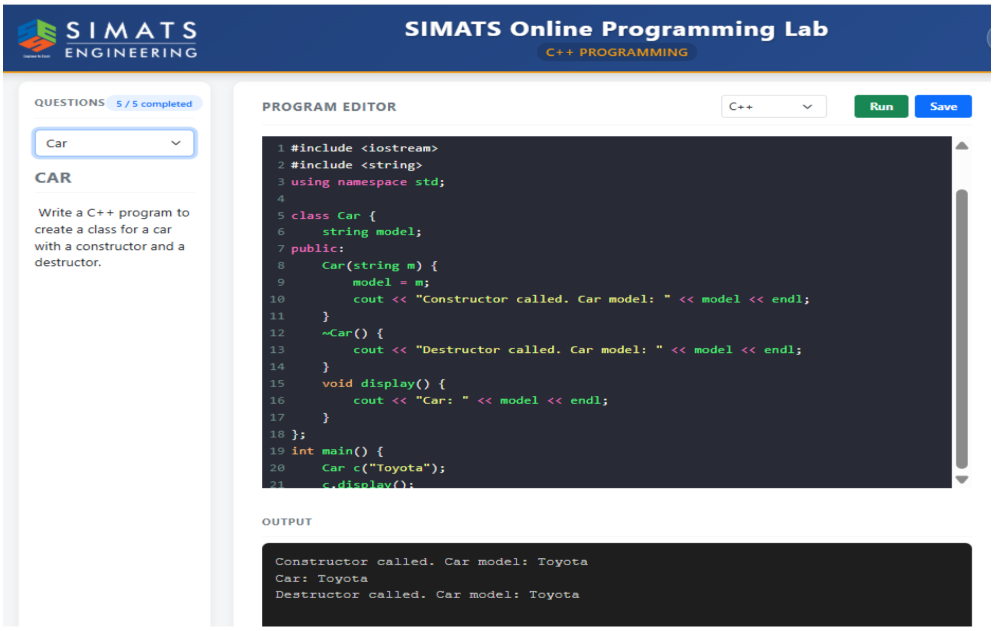
```
#include <iostream>
using namespace std;

class Rectangle {
private:
   double length;
   double width;

public:
   Rectangle(double l, double w) {
      length = l;
      width = w;
      cout << "Constructor called: Rectangle created with length = "
          << length << " and width = " << width << endl;
   }

   double area() {
      return length * width;
   }

   double perimeter() {
      return 2 * (length + width);
   }

   ~Rectangle() {
      cout << "Destructor called: Rectangle object destroyed." << endl;
   }
```

```
};

int main() {
    double l, w;

    cout << "Enter length and width of the rectangle: ";
    cin >> l >> w;

    Rectangle rect(l, w);

    cout << "Area of rectangle: " << rect.area() << endl;
    cout << "Perimeter of rectangle: " << rect.perimeter() << endl;

    return 0;
}
```



4.**Creating a class for a car with a constructor and destructor**

```
#include <iostream>
#include <string>
using namespace std;

class Car {
string model;
public:
Car(string m) {
model = m;
cout << "Constructor called. Car model: " << model << endl;
}
~Car() {
cout << "Destructor called. Car model: " << model << endl;
}
void display() {
cout << "Car: " << model << endl;
}
};

int main() {
Car c("Toyota");
c.display();
return 0;
}
```



5.**Creating a class for a bank with a constructor and destructor**

```
#include <iostream>
#include <string>
using namespace std;

class BankAccount {
private:
string name;
int accNo;
double balance;

public:
// Constructor (used when object is created)
BankAccount(string n, int a, double b) {
name = n;
accNo = a;
balance = b;
}

// Destructor (runs automatically when object is destroyed)
~BankAccount() {
cout << "Destructor called for account: " << accNo << endl;
}

void display() {
cout << "Account Holder: " << name
<< "\nAccount Number: " << accNo
<< "\nBalance: " << balance << endl;
}
};

int main() {
BankAccount acc("Supriya", 100005, 5000.0); // constructor called here
acc.display(); // show data
return 0; // destructor called here
}
```

QUESTIONS 5 / 5 completed

Bank

BANK

Write a C++ program to create a class for a bank account with a constructor and a destructor.

PROGRAM EDITOR

C++ Run Save

```cpp
    double balance;
public:
    BankAccount(string n, int a, double b) {
        name = n;
        accNo = a;
        balance = b;
    }
    ~BankAccount() {
        cout << "Destructor called for account: " << accNo << endl;
    }
    void display() {
        cout << "Account Holder: " << name
             << "\nAccount Number: " << accNo
             << "\nBalance: " << balance << endl;
    }
};
int main() {
    BankAccount acc("Supriya", 100005, 5000.0);
    acc.display();
    return 0;
}
```

OUTPUT

```
Account Holder: Supriya
Account Number: 100005
Balance: 5000
Destructor called for account: 100005
```